

Lecture 7: Introduction to Supercomputers and Performance Metrics

Objective:

To introduce the architecture of supercomputers and explain key performance metrics used in High-Performance Computing (HPC).

Duration: 1 Hour

Learning Outcomes

By the end of this lecture, students will be able to:

- Describe the components of a supercomputer
- Understand examples of world-class supercomputers
- Explain key factors like scalability, reliability, and energy efficiency
- Interpret HPC performance metrics: FLOPS, latency, bandwidth, speedup, and efficiency

1. Introduction to Supercomputers

What is a Supercomputer?

A **supercomputer** is a high-speed computing system designed to perform complex scientific, engineering, or simulation problems that require immense computational power.

Components of a Supercomputer

Component	Description
Nodes	Basic computational units (each node may have CPUs/GPUs)
Interconnects	High-speed networks linking nodes (e.g., InfiniBand)
Storage	Large, fast-access storage (parallel/distributed file systems like Lustre or GPFS)

Examples of Famous Supercomputers

1. Fugaku (Japan – RIKEN Center)

- **Architecture:** ARM-based Fujitsu A64FX processors
- **Peak Performance:** ~442 PFLOPS
- **Use Case:** Drug discovery, weather forecasting, pandemic modeling

2. Summit (USA – Oak Ridge National Lab)

- **Architecture:** IBM POWER9 CPUs + NVIDIA Tesla V100 GPUs
- **Peak Performance:** ~200 PFLOPS
- **Use Case:** Climate modeling, genomics, physics simulations

IN Top Supercomputers in India

1. PARAM Siddhi-AI

- **Developed by:** C-DAC (Centre for Development of Advanced Computing) under NSM
- **Location:** C-DAC, Pune
- **Processor:** NVIDIA DGX SuperPOD with Intel Xeon CPUs + NVIDIA V100 GPUs
- **Peak Performance:** 5.267 PFLOPS
- **Global Rank (Top500, 2023):** ~63
- **Use Case:** AI applications, drug discovery, genomics, weather forecasting, pandemic modeling

2. Pratyush

- **Developed by:** Indian Institute of Tropical Meteorology (IITM), Pune
- **Processor:** Intel Xeon Broadwell E5-2695 + NVIDIA Tesla K40 GPUs
- **Peak Performance:** 4.0 PFLOPS

- **Use Case: Climate and weather forecasting, monsoon prediction**

3. Mihir

- **Location: National Centre for Medium Range Weather Forecasting (NCMRWF), Noida**
- **Processor: Cray XC40 with Intel Xeon Broadwell**
- **Peak Performance: 2.8 PFLOPS**
- **Use Case: Numerical weather prediction, disaster management systems**

4. PARAM Shivay

- **Installed at: IIT (BHU), Varanasi**
- **Peak Performance: 833 TFLOPS**
- **Purpose: Supporting research and academic computing under National Supercomputing Mission (NSM)**

2. Key Factors in Supercomputing

Scalability

- Ability of a system to maintain efficiency as the number of processors increases
- Critical for applications run on thousands of processors

Reliability

- Ensures system uptime and fault tolerance
- Redundancy, error correction, and failover mechanisms are important

Energy Efficiency

- Measured as **performance per watt**
- Example metric: GFLOPS/Watt

- **Green500 list** ranks the most energy-efficient supercomputers

3. HPC Performance Metrics

a. FLOPS (Floating Point Operations Per Second)

- **Definition:** Number of floating-point operations a computer can perform per second
- **Example:**
 - 1 Gigaflops = 10^9 operations/sec
 - 1 Teraflops = 10^{12} operations/sec
- Used to measure **raw computational power**

b. Latency

- Time taken to transfer a message from source to destination
- Lower latency = faster response
- Important in communication between nodes

c. Bandwidth

- Amount of data transferred per unit time (e.g., GB/s)
- Higher bandwidth = better data throughput
- Key in memory and I/O-intensive applications

d. Speedup

- **Formula:**

$$\text{Speedup} = \frac{T_{\text{serial}}}{T_{\text{parallel}}}$$

where T_{serial} = time taken by single processor
and T_{parallel} = time taken by parallel processors

- **Example:** If a task takes 10s serially and 2s in parallel $\rightarrow$ speedup = $5\times$

e. Efficiency

- Measures how well parallel resources are used
- **Formula:**

$$\text{Efficiency} = \frac{\text{Speedup}}{\text{Number of Processors}} \times 100\%$$

- **Example:**
Speedup = 5 using 10 processors $\rightarrow$ Efficiency = 50%

Classroom Activities

♦ Activity 1: Match the Metric

- Match terms (FLOPS, latency, speedup) with their correct definitions

♦ Activity 2: Discussion

- Question: Why is energy efficiency a bigger concern today than in the past?

♦ Activity 3: Solve an Example

- If a program runs in 60s on 1 core and 15s on 4 cores:

- Speedup = ?
- Efficiency = ?

Solution:

- Speedup = $60 / 15 = 4$
- Efficiency = $(4 / 4) \times 100\% = 100\%$

 Summary

Topic	Key Takeaways
Supercomputer	Very fast, large-scale system used for research & simulation
Components	Nodes, Interconnects, Storage
Key Factors	Scalability, Reliability, Energy Efficiency
Performance Metrics	FLOPS (power), Latency (delay), Bandwidth (speed), Speedup (gain), Efficiency (usage)

 Homework/Assignment

Task:

Research any supercomputer from the **Top500** list. Write a 250-word report that includes:

- Name and country
- Architecture
- Peak FLOPS
- Energy efficiency rating
- A real-world application it's used for

Submission Deadline: Before the next lecture